

required. The flow rate of methacrylic acid production will be approximately be the same. Using patent no. CN114057571A as a reference, we get the value of isobutylene molar flow rate as 41.75 kmol/h. The calculations done for this is as follows -

$$F_{\text{MAA}} = F_{\text{MAL}} \times \text{yield of MAA} = F_{\text{MAL}} \times 0.663$$

$$F_{\text{MAL}} = 33.23 \text{ kmol/h}$$

$$F_{\text{MAL}} = F_{\text{IB}} \times \text{yield of MAL} = F_{\text{IB}} \times 0.796$$

$$F_{\text{IB}} = 41.75 \text{ kmol/h.}$$

Where, F_x is the molar flow rate of the species “x”,

MAL is abbreviation of Methacrolein

MAA is abbreviation of Methacrylic acid.

Now from literature, it is seen that the ideal range for the molar ratio of IB to Oxygen to steam is 1: 2: 1.5. Using this composition, we can easily calculate the inlet flow rate to reactor R201 and its composition. This stream is labelled as stream 4 in the Process flow diagram.

Table 17: Stream composition of inlet to the first reactor

Stream 4	mol%	kmol/h	mw	kg/h
IB	8.63	41.75	56	2,338.00
O2	17.25	83.50	32	2,672.00
N2	61.18	296.05	28	8,289.27

steam	12.94	62.63	18	1,127.25
Total	100	483.92		14,426.52

The mass balance of the outlet stream from the reactor and its succeeding units has been carried out and found out based on the compositions mentioned in the patent CN114057571A. For the species whose molar composition in a stream was unknown, a basic elemental balance was used to find out the same. For each stream it was ensured that the mass in was equal to the mass out.

Given below is the list of the all the units in the plant

Table 18: List of the units in the plant

Equipment	Description	Type
R201	Oxidation reactor	Multi tubular packer bed
R202	Oxidation reactor	Multi tubular packer bed
C301	Quencher	Direct contact using water
C203	Crystallizer	Batch crystallizer using cooling water
Filter		To filter out TPA crystals
T204	Extraction column	Packed column
T302	Distillation column	Plate column
T303	Distillation column	Plate column
T304	Distillation column	packed

T204	Distillation column	packed
R305	Esterification reactor	Cstr (gas-liq)
T306	Distillation column	packed

6.1 Reactor system material balance

For the first oxidation reactor **R201**, the feed is Isobutene, Steam and air mixture which is initially heated to 350 °C and sent to the reactor. The reactor is a multi-tubular packed bed reactor which is said to operate isothermally. The coolant used is a thermic salt.

The conversion of Isobutene is 99.4% and the selectivity to methacrolein is 80%. Using these values, we can calculate the amount of Methacrolein that will be formed. Let molar flow rate of isobutene be denoted by F_{IB0} and the molar flow rate of methacrolein be F_{MAC} .

So,

$$F_{MAC} = 0.994 \times 0.8 \times F_{IB}$$

$$F_{MAC} = 0.994 \times 0.8 \times 41.75$$

$$F_{MAC} = 33.23 \text{ kmol/h}$$

Patent CN114057571A mentions the composition of the reactor outlet but the value of steam and oxygen was not given. So using the elemental balance for carbon and oxygen, the outlet flow rate of steam and water was also calculated. Nitrogen is an inert gas so the moles of nitrogen in and moles out will be the same. Table 19 gives the outlet of the reactor 1.

Table 19: Stream Composition of outlet of reactor 201

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate (kg/h)	wt%
IB	0.03	0.17	56.00	3.61	0.03

MAC	6.84	33.23	70.00	2,326.31	16.13
CO	1.05	5.12	28.00	143.46	0.99
MAA	0.30	1.45	86.00	125.04	0.87
Acetic acid	0.30	1.45	60.00	87.24	0.60
Acetone	0.17	0.83	58.00	48.19	0.33
Acetalde	0.30	1.45	44.00	63.97	0.44
Acrolein	0.06	0.28	56.00	15.51	0.11
Formaldehyde	0.04	0.21	30.00	6.23	0.04
Acrylic acid	0.04	0.21	72.00	14.95	0.10
O2	4.99	24.23	32.00	775.44	5.38
N2	60.91	296.05	28.00	8,289.27	57.46
CO2	0.87	4.24	44.00	186.55	1.29
Steam	22.39	108.82	18.00	1,958.78	13.58
Formic acid	1.70	8.25	46.00	379.55	2.63
Total	100.00	486.00		14,426.52	100.00

Similarly, for the second oxidation reactor **R202**, the conversion of methacrolein is 78 % and the selectivity to methacrylic acid is 85 %. So the molar flow rate of methacrylic acid FMAA will be given by -

$$F_{MAA} = 0.85 \times 0.78 \times F_{MAC}$$

$$F_{MAA} = 0.85 \times 0.78 \times 33.23$$

$$F_{MAA} = 23.81 \text{ kmol/h}$$

Again using the compositions given in the patent, the outlet of the reactor R202 was also calculated. The makeup air that was to be added to the reactor, was calculated by elemental balance of Nitrogen, Oxygen and Carbon. The stream composition of the reactor outlet is given below. The mass flow rate of the makeup air is 13,127.27 kg/h.

Table 20: Stream composition of outlet of Reactor R202

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate (kg/h)	wt%
MAC	0.70	7.31	70.00	511.79	1.86
CO	1.00	10.44	28.00	292.45	1.06
MAA	2.28	23.81	86.00	2,047.99	7.43
Acetic ac	0.31	3.24	60.00	194.27	0.71
acetone	0.12	1.25	58.00	72.69	0.26
acetaldehyde	0.21	2.19	44.00	96.51	0.35
acrolein	0.04	0.42	56.00	23.40	0.08
formaldehyde	0.03	0.31	30.00	9.40	0.03
acrylic acid	0.03	0.31	72.00	22.56	0.08

oxygen	6.50	67.89	32.00	2,172.49	7.88
steam	18.98	198.28	18.00	3,569.02	12.95
CO ₂	0.69	7.25	44.00	318.83	1.16
N ₂	62.29	650.59	28.00	18,216.55	66.11
Impurity	0.003	0.035	166.00	5.85	0.02
Total		1,044.47		27,553.80	

6.2 Quencher

Before sending the outlet gas from the reactor to the quencher, it is cooled to 230 °C and then sent to the quencher. The quenching occurs at 80 °C, this separates methacrolein, acrolein and the lighter carbon compounds in the gas phase while the liquid phase will contain some amount of Methacrolein dissolved. The liquid phase will also contain Methacrylic acid, Methacrolein, Impurity which is Terephthalic acid, acrylic acid, Acetic acid and large quantity of water. For all the separation steps after the reactor, 1000 ppm of a polymerisation inhibitor (in our case MEHQ) needs to be sprayed throughout the columns to avoid the formation of polymers. The gas stream composition is given in Table 21. The gas stream is sent to the flare for combustion.

Table 21: Gas stream composition (outlet of quencher)

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate(kg/h)	Wt%
MAC	0.96	7.14	70.00	500.09	2.30
CO	1.40	10.44	28.00	292.45	1.35

Acetone	0.17	1.25	58.00	72.69	0.33
acetaldehyde	0.29	2.19	44.00	96.51	0.44
acrolein	0.06	0.42	56.00	23.40	0.11
oxygen	9.08	67.89	32.00	2,172.49	10.01
formaldehyde	0.04	0.31	30.00	9.40	0.04
CO2	0.97	7.25	44.00	318.83	1.47
N2	87.04	650.59	28.00	18,216.55	83.94
Total		747.49		21,702.40	

Similarly the composition of the liquid stream is given in

Table 22: Liquid stream composition (outlet of quencher)

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate (kg/h)	Wt%
MAA	10.54	23.81	86.00	2,047.99	35.00
MAC	0.07	0.17	70.00	11.70	0.20
Impurity	0.02	0.04	166.00	5.85	0.10

acrylic acid	0.14	0.31	72.00	22.56	0.39
acetic acid	1.43	3.24	60.00	194.27	3.32
water	87.79	198.28	18.00	3,569.02	60.99
Total		225.85		5,851.40	100.00

The liquid stream is then sent to the crystallizer to crystal out TPA at 13 °C. The entire TPA is crystallised out and the liquid stream outlet of the crystallizer is sent to an extraction column.

6.3 Extraction column

Since the compounds have the ability to polymerize, there is no data available for the Liquid-liquid equilibrium data. So for the extraction column, a simulation was run on DWSIM where the interaction parameters were calculated using UNIFAC and the model used for LLE was NRTL. Solvent chosen for extraction was n-Butyl Acetate (NBA) which has the tendency to go into the organic phase and has low solubility in water. The simulation report is attached in Appendix A.

The amount of solvent requires was 8000 kg/h. Table 23 is the stream composition of the extract phase.

Table 23: Stream composition of extract phase

Species	Mass flow rate (kg/h)	MW	Molar flow rate (kmol/h)	wt%	mol%
MAA	2,028.52	86.00	23.59	18.40	15.82
NBA	7,800.10	116	67.24	70.75	45.11
Water	986.65	18	54.81	8.95	36.77

Acrylic acid	22.57	72.00	0.31	0.20	0.21
Acetic acid	186.54	60.00	3.11	1.69	2.09
MAC	11.70	70.00	0.17	0.11	0.11
Total	11,024.38		149.07		

We can see that most of the methacrylic acid is extracted along with the other organics and some amount of water. However the solvent extracted is also almost equal to the solvent fed. This will make the distillation of this stream to purify the methacrylic acid more rigorous and energy intensive.

Table 24: Stream composition of the raffinate phase

Species	Mass flow rate (kg/h)	MW	Molar flow rate (kmol/h)	wt%	mol%	
MAA	19.47	86.00		0.23	0.69	0.15
NBA	199.90	86		2.32	7.12	1.59
Water	2,582.37	18		143.47	91.92	98.17
Acrylic acid	0.00	72.00		0.00	0.00	0.00
Acetic acid	7.73	60.00		0.13	0.28	0.09
MAC	0.00	70.00		0.00	0.00	0.00
Total	2,809.47			146.14		

From Table 24, we can see that the distribution coefficient is 27. The raffinate stream is also almost water and can be sent to the ETP.

Chapter 7: Energy balance

The values of specific heat and the enthalpy of formation were taken from Yaws' Handbook of physical Properties. Both were functions of temperature. Based on these values, energy balance was done for all the three reactors.

The enthalpy of formation (H_f) is given by

$$H_f = A + BT + CT^2,$$

where H_f is in kJ/mol, T is in K and the component is in gas phase.

Table 25: Regression coefficients for Enthalpy of Formation based on Temperature

Species	A	B	C
CO	-112.19	8.12E-03	-8.04E-06
CO2	-393.422	1.50E-04	-1.39E-06
Formaldehyde	-109.671	-2.32E-02	7.87E-06
acetic acid	-422.584	-4.84E+00	2.34E-05
Acetaldehyde	-154.122	-4.72E-02	2.03E-05
Acrolein	-66.702	-5.95E-02	3.87E-05
Acrylic acid	-325.038	-4.41E-02	2.09E-05
Acetone	-199.175	-7.15E-02	3.25E-05
Methacrolein	-90.882	-8.68E-02	5.28E-05
Methacrylic acid	-430.04	-4.75E-02	3.47E-04
Impurity (TPA)	-688.693	-1.13E-01	4.93E-05
Water	-238.41	-1.23E-02	2.77E-06
Ethylene Oxide	-38.88	-5.40E-02	2.56E-05
Isobutylene	4.126	-8.16E-02	3.66E-05

Similarly the values of C_p were calculated for the component in gas phase

$$C_p = A + BT + CT^2 + DT^3 + ET^4$$

where C_p is in J/molK and T is in K.

Table 26: Regression coefficients for Specific heat based on Temperature

Species	A	B	C	D	E
MAL	14.506	1.59E-01	3.12E-04	-4.21E-07	1.42E-10
CO	29.556	-6.58E-03	2.01E-05	-1.22E-08	2.67E-12
MAA	-28.131	5.47E-01	-5.39E-04	2.86E-07	-6.09E-11
Acetic ac	34.85	3.76E-02	2.83E-04	-3.08E-07	9.26E-11
acetone	35.918	9.39E-02	1.87E-04	-2.16E-07	6.32E-11
acetaldehyd e	34.14	4.00E-02	1.56E-04	1.64E-07	4.72E-11
acrolein	109.243	-5.10E-01	1.71E-03	-1.81E-06	6.60E-10
formaldehy de	34.428	-2.98E-02	1.51E-04	-1.27E-07	3.39E-11
acrylic acid	7.755	2.94E-01	-2.09E-04	7.16E-08	-9.10E-12
oxygen	31.32234	-2.02E-02	5.79E-05	-3.65E-08	7.37E-15
steam	30.092	6.83E-03	6.79E-06	-2.53E-09	8.21E-14
CO2	27.437	4.23E-02	-1.96E-05	4.00E-09	-2.99E-13